

Grade 7 Accelerated
Unit 3
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

1. What is the value of 10^{-16} , written as a fraction?

$$\frac{1}{10,000,000,000,000,000}$$

2. Complete the statements.

The expression $\frac{10^{-17}}{10^{-5}}$ can be written can as 10^{-12} . Another way to write the expression $\frac{10^{-17}}{10^{-5}}$ using positive exponents only is $\frac{1}{10^{12}}$ or $\frac{10^5}{10^{17}}$.

3. Complete the table. Write each mathematical statement in an equivalent form by applying once or more laws of exponents. Then, write the equivalent decimal value.

Mathematical Statement	Value Written as a Power of 10	Decimal Equivalent
$(10^4) - 4 \times 10^{25}$	Not possible	$-40,000,000,000,000,000,000,000$
$(\frac{1}{10^{-5}})^3$	10^{15}	$1,000,000,000,000,000$
$\frac{10^{-14}}{10^{-9}} \cdot 10^6$	$10^{-5} \cdot 10^6 = 10$	10

Express each number in scientific notation.

4. The diameter of a human hair is about 67 hundred-thousandths of an inch.

$$6.7 \times 10^{-4}$$

5. 0.0000000049

$$4.9 \times 10^{-9}$$

Express each number in standard form.

6. A cell's diameter 0.0174×10^{-4}

$$0.00000174$$

7. 9×10^{-6}

$$0.000009$$

8. Complete the table.

Scientific Notation	Standard Form
2.6×10^{-8}	0.000000026
5.2×10^{-3}	0.0052

9. Complete the statements using the table in question 8.

o 5.2 is 2 times ☐ smaller ☒ larger than 2.6.

o 10^{-3} is 10^5 times ☐ smaller ☒ larger than 10^{-8} .

o The value 5.2×10^{-3} is 2×10^5 or 200,000 times ☐ smaller ☒ larger than 0.000000026.